DSA0404- FUNDEMENTALS OF DATA SCIENCE **M Poojitha Reddy**

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CO3- SOLVED PROBLEMS

Code of Conduct:

I ___**M Poojitha Reddy**____ (Name / Reg No) certify that this submission is my original work

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Signature of the student:

M Poojitha Reddy

1. **Find mean, median, mode and range for: 12, 15, 18, 15, 20, 22, 15, 25.**

Given data:

12, 15, 18, 15, 20, 22, 15, 25

Step 1: Arrange in ascending order

12, 15, 15, 15, 18, 20, 22, 25

Number of observations:

n = 8

Step 2: Mean

Formula:

Mean = Sum of all observations / Number of observations

Sum:

12 + 15 + 18 + 15 + 20 + 22 + 15 + 25

= **142**

Therefore,

Mean = 142 / 8

= **17.75**

Step 3: Median

There are 8 observations, so it is an even number.

Median = Average of the 4th and 5th values.

4th value = 15
5th value = 18

Median = (15 + 18) / 2

= 33 / 2

= **16.5**

Step 4: Mode

Mode is the value that occurs most frequently.

15 occurs **3 times**.

Therefore,

Mode = 15

Step 5: Range

Formula:

Range = Maximum value − Minimum value

= 25 − 12

= **13**

Final Answer

- **Mean = 17.75**
- **Median = 16.5**
- **Mode = 15**
- **Range = 13**

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2. Construct a frequency distribution for: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.

Given:

2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4

Step 1: Identify unique values

The values are:

2, 3, 4, 5, 6

Step 2: Count each value

Value	Frequency
2	3

Value	Frequency
3	3
4	3
5	2
6	1

Step 3: Verify

Total frequency:

3 + 3 + 3 + 2 + 1 = **12**

There are 12 observations in the original data.

Therefore, the frequency distribution is correct.

Final Answer

2 → 3 times
3 → 3 times
4 → 3 times
5 → 2 times
6 → 1 time

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3. Find Q1, Q2, Q3 and IQR for: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30.

Given data:

5, 8, 10, 12, 15, 18, 20, 22, 25, 30

Already arranged in ascending order.

Number of observations:

n = 10

We divide the data into two halves.

Lower half:

5, 8, 10, 12, 15

Upper half:

18, 20, 22, 25, 30

Step 1: Find Q2

Q2 is the median of the complete dataset.

Since n = 10:

Q2 = Average of 5th and 6th values

5th value = 15
6th value = 18

Q2 = (15 + 18) / 2

= **16.5**

Step 2: Find Q1

Q1 is the median of the lower half:

5, 8, 10, 12, 15

The middle value is **10**.

Therefore:

Q1 = 10

Step 3: Find Q3

Q3 is the median of the upper half:

18, 20, 22, 25, 30

The middle value is **22**.

Therefore:

Q3 = 22

Step 4: Find IQR

Formula:

IQR = Q3 − Q1

= 22 − 10

= **12**

Final Answer

- **Q1 = 10**
- **Q2 = 16.5**

- **Q3 = 22**
- **IQR = 12**

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4. Detect outliers using the IQR method for: 10, 12, 13, 15, 16, 17, 18, 20, 45.

Given:

10, 12, 13, 15, 16, 17, 18, 20, 45

Already sorted.

Number of observations:

n = 9

Step 1: Find Q2

Since there are 9 observations, the middle value is the 5th value.

Q2 = **16**

Now divide the remaining values:

Lower half:

10, 12, 13, 15

Upper half:

17, 18, 20, 45

Step 2: Find Q1

Lower half:

10, 12, 13, 15

There are 4 values.

Q1 = Average of 2nd and 3rd values.

Q1 = (12 + 13) / 2

= **12.5**

Step 3: Find Q3

Upper half:

17, 18, 20, 45

Q3 = Average of 2nd and 3rd values.

Q3 = (18 + 20) / 2

= **19**

Step 4: Find IQR

IQR = Q3 − Q1

= 19 − 12.5

= **6.5**

Step 5: Find lower boundary

Formula:

Lower Boundary = Q1 − 1.5 × IQR

= 12.5 − 1.5(6.5)

= 12.5 − 9.75

= **2.75**

Step 6: Find upper boundary

Formula:

Upper Boundary = Q3 + 1.5 × IQR

= 19 + 1.5(6.5)

= 19 + 9.75

= **28.75**

Step 7: Identify outlier

Any value:

- Less than 2.75 → outlier
- Greater than 28.75 → outlier

The value **45 > 28.75**.

Therefore:

45 is an outlier.

Final Answer

Outlier = 45

5. Determine the skewness (Asymmetry) of: 5, 6, 7, 7, 8, 8, 9, 20.

Given:

5, 6, 7, 7, 8, 8, 9, 20

Step 1: Find the mean

Add all values:

5 + 6 + 7 + 7 + 8 + 8 + 9 + 20

= **70**

Number of observations:

n = 8

Mean:

70 / 8 = **8.75**

Step 2: Find median

There are 8 observations.

Median = Average of 4th and 5th values.

4th value = 7
5th value = 8

Median:

(7 + 8) / 2 = **7.5**

Step 3: Find mode

7 occurs twice.

8 occurs twice.

Therefore, there are **two modes: 7 and 8**.

Step 4: Compare Mean and Median

Mean = **8.75**

Median = **7.5**

So:

Mean > Median

The extreme value **20** pulls the mean toward the right.

Therefore, the distribution is:

Positively skewed / Right-skewed

Final Answer

The data is positively skewed (right-skewed).

Reason:

Mean > Median, mainly because of the large value **20**.

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6. **A uniformly distributed variable ranges from 10 to 50. Find P(20<X<35).**

A uniformly distributed variable ranges from:

10 to 50

Find:

P(20 < X < 35)

Step 1: Understand uniform distribution

For a continuous uniform distribution:

P(a < X < b) = Length of desired interval / Total length

Step 2: Find total interval

Total range:

50 − 10 = **40**

Step 3: Find desired interval

From 20 to 35:

35 − 20 = **15**

Step 4: Calculate probability

P(20 < X < 35)

= 15 / 40

= 0.375

Therefore:

P(20 < X < 35) = 0.375

As percentage:

0.375 × 100 = **37.5%**

Final Answer

P(20 < X < 35) = 0.375 = 37.5%

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7. **A uniformly distributed variable ranges from 10 to 50. Find P(20<X<35).**

Given:

Mean:

μ = 100

Standard deviation:

σ = 15

Value:

X = 130

Step 1: Write the formula

Z-score formula:

$Z = (X - \mu) / \sigma$

Step 2: Substitute values

$Z = (130 - 100) / 15$

Step 3: Calculate

$Z = 30 / 15$

= **2**

Final Answer

Z = 2

Interpretation

The value 130 is **2 standard deviations above the mean**.

8.For the same normal distribution, find P(X<130).

We already found:

Z = 2

So we need:

P(X < 130) = P(Z < 2)

Step 1: Look at the standard normal table

For:

Z = 2.00

The standard normal table gives:

P(Z < 2.00) = 0.9772

Step 2: Convert to percentage

0.9772 × 100

= **97.72%**

Final Answer

P(X < 130) = 0.9772

or

P(X < 130) = 97.72%

Interpretation

Approximately **97.72% of observations are below 130**.

9.Given a dataset with one extreme value, compare the mean and median before and after outlier treatment.

Let's take a simple dataset:

10, 12, 13, 14, 15, 16, 100

Here, **100 is an extreme outlier**.

Before Outlier Treatment

Step 1: Find mean

Sum:

10 + 12 + 13 + 14 + 15 + 16 + 100

= **180**

Number of observations:

n = 7

Mean:

180 / 7

= **25.71**

Step 2: Find median

There are 7 observations.

The middle value is the 4th value.

Median = **14**

So before treatment:

Mean = 25.71

Median = 14

Notice how the mean has become very large because of 100.

After Outlier Treatment

Remove the extreme value **100**.

New dataset:

10, 12, 13, 14, 15, 16

Step 1: Find new mean

Sum:

10 + 12 + 13 + 14 + 15 + 16

= **80**

Number of observations:

n = 6

Mean:

80 / 6

= **13.33**

Step 2: Find new median

There are 6 observations.

Median = Average of 3rd and 4th values.

3rd value = 13
4th value = 14

Median:

(13 + 14) / 2

= **13.5**

Comparison

Measure	Before treatment	After treatment
Mean	25.71	13.33
Median	14	13.5

Conclusion

The outlier **100 greatly increased the mean**.

The median changed only slightly.

Therefore:

Mean is more sensitive to outliers, while median is more resistant to outliers.

10.Perform a complete EDA on a small student-mark dataset and report central tendency, dispersion, distribution, outliers and skewness.

Let's take this small dataset:

Student Marks:

55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 95

We will perform EDA step by step.

Step 1: Arrange the data

The data is already arranged:

55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 95

Number of students:

n = 11

A. Central Tendency

We find:

1. Mean
2. Median
3. Mode

Step 2: Calculate Mean

Formula:

Mean = Sum of values / Number of values

Add all marks:

55 + 60 + 62 + 65 + 68 + 70 + 72 + 75 + 78 + 80 + 95

= **780**

Number of students = 11

Mean:

780 / 11

= **70.91**

Therefore:

Mean = 70.91 marks

Step 3: Calculate Median

There are 11 observations.

Since n is odd:

Median = (n + 1) / 2 position

= (11 + 1) / 2

= 12 / 2

= **6th position**

The 6th value is:

70

Therefore:

Median = 70

Step 4: Calculate Mode

Look at the values:

55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 95

Every value occurs only once.

Therefore:

There is no mode.

B. Dispersion

We can calculate:

1. Range
2. Variance
3. Standard deviation
4. IQR

Step 5: Calculate Range

Formula:

Range = Maximum − Minimum

Maximum = 95

Minimum = 55

Range:

95 − 55

= **40**

Therefore:

Range = 40 marks

Step 6: Find Q1 and Q3

Data:

55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 95

Median = 70.

Lower half:

55, 60, 62, 65, 68

Middle value = **62**

Therefore:

Q1 = 62

Upper half:

72, 75, 78, 80, 95

Middle value = **78**

Therefore:

Q3 = 78

Step 7: Calculate IQR

Formula:

IQR = Q3 − Q1

= 78 − 62

= **16**

Therefore:

IQR = 16 marks

C. Detect Outliers

Use the IQR method.

Step 8: Lower boundary

Formula:

Lower Boundary = Q1 − 1.5 × IQR

= 62 − 1.5(16)

= 62 − 24

= **38**

Step 9: Upper boundary

Formula:

Upper Boundary = Q3 + 1.5 × IQR

= 78 + 1.5(16)

= 78 + 24

= **102**

Therefore, acceptable values lie between:

38 and 102

Our smallest value = 55

Our largest value = 95

Both are within this range.

Therefore:

There are no outliers.

D. Distribution / Shape

Now compare mean and median.

Mean = **70.91**

Median = **70**

Therefore:

Mean > Median

This indicates a slight **positive/right skew**.

Why?

The value **95** is relatively high compared with most of the other marks, so it pulls the mean slightly toward the right.

Therefore:

Distribution = Slightly positively skewed.

E. Variance and Standard Deviation

For a simple EDA, we can also calculate the spread using standard deviation.

Mean ≈ **70.91**

We calculate each deviation from the mean and square it.

Marks (x)	x − Mean	(x − Mean)²
55	-15.91	253.13
60	-10.91	119.01
62	-8.91	79.39
65	-5.91	34.92
68	-2.91	8.46
70	-0.91	0.83
72	1.09	1.19
75	4.09	16.74
78	7.09	50.28
80	9.09	82.64
95	24.09	580.32

Sum of squared deviations ≈ **1226.91**

Population variance

Formula:

Variance = $\Sigma(x - \mu)^2 / n$

= 1226.91 / 11

≈ **111.54**

Standard deviation

Formula:

SD = $\sqrt{\text{Variance}}$

= $\sqrt{111.54}$

≈ **10.56**

Therefore:

Standard deviation ≈ 10.56 marks